INSTITUTE OF MATHEMATICAL SCIENCES, FACULTY OF SCIENCE, UNIVERSITI MALAYA
SEMESTER 2, SESSION 2025/2026
TUTORIAL 9

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1. For each differential equation, identify all singular points (if any), and classify each as regular or irregular.
   (a) $y'' - x^3y' + xy = 0.$
   (b) $(x^2 - 3)y'' - y = 0.$
   (c) $e^x y'' - (x^2 - 1)y' + 2xy = 0.$

2. Find two linearly independent series solutions $y_1(x)$ and $y_2(x)$ of the following differential equations.
   (a) $y'' - y = 0.$
   (b) $y'' + 2y' = 0.$

3. Consider the following differential equation.

$$y'' - xy = 0.$$

   (a) Determine whether $x_0 = 0$ is a singular point or an ordinary point?
   (b) Then, find the general series solution for the above differential equation centered at $x_0 = 0$.

4. Verify that x_0 = 0 is a regular singular point for the following differential equation

$$2xy'' + y' + x^3y = 0$$

   Use the method of Frobenius to find the general solution $y(x) = Ay_1(x) + By_2(x)$.